NLP & MACHINE LEARNING

Resume Classification

Automated talent screening and role categorization using Natural Language Processing.

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THE PROBLEM

The HR Bottleneck

- **Manual Screening:** Recruiters spend hours parsing unstructured text.
- **Human Bias:** Inconsistent evaluation criteria across reviewers.
- **Format Chaos:** Infinite variations in layout, vocabulary, and file types.



System Objectives

Text Extraction

Robust parsing of raw PDFs and DOCX files.

NLP Cleaning

Lemmatization, stop-word removal, and tokenization.

Role Classification

Map unstructured text to predefined job categories.

NLP Stack



Python



spaCy / NLTK



Scikit-Learn



PyPDF2

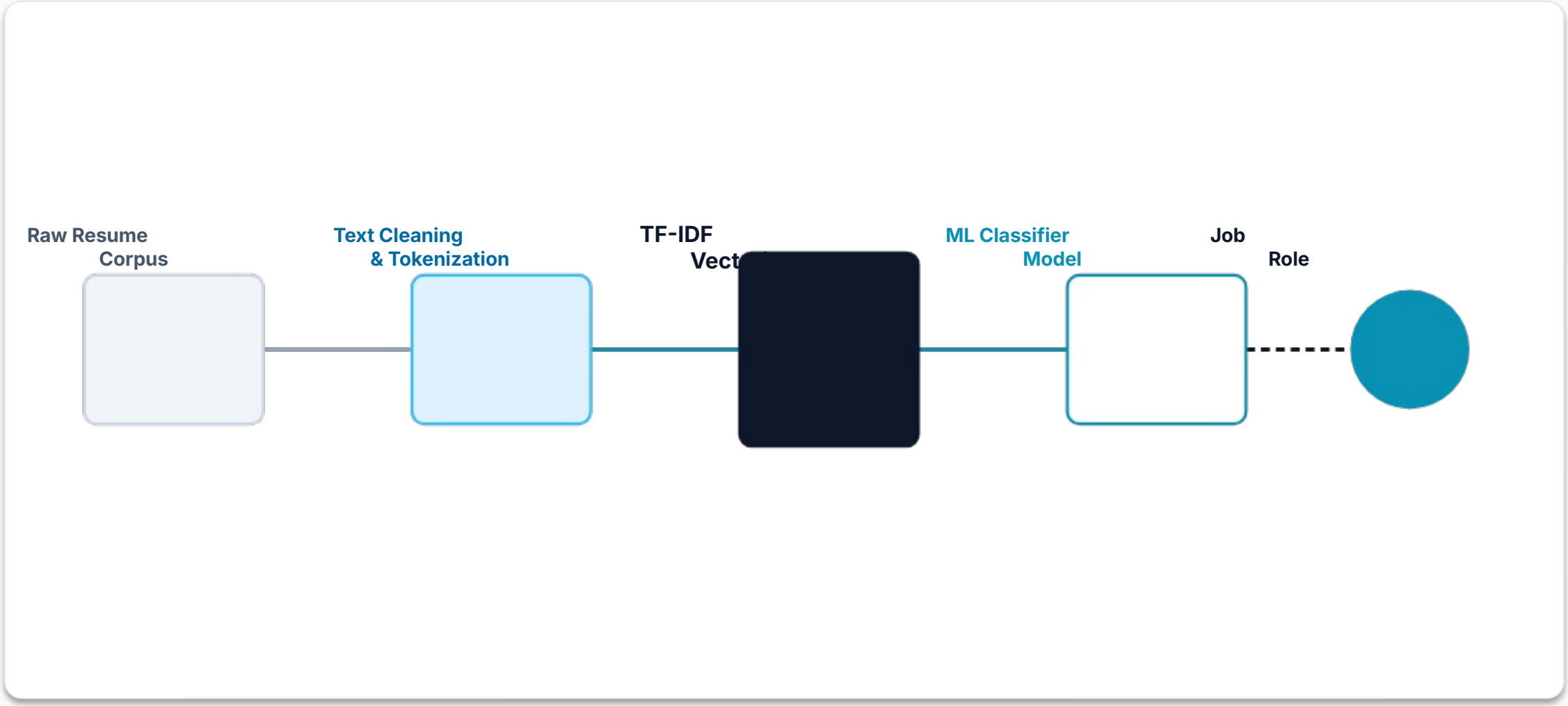


Pandas



Regex

Classification Architecture



NLP Pipeline

1. Ingest

Extract string from PDF.

2. Clean

Regex removes emails & URLs.

3. Vectorize

Convert tokens to numeric matrices.

4. Classify

Model assigns domain category.

Dataset & Preprocessing

The Corpus

- Kaggle / Open-source resume dataset.
- Labeled by domain (IT, Finance, HR).
- Class balancing applied via SMOTE.

Text Normalization

- Stopword removal (the, is, and).
- Lemmatization (running → run).
- Lowercase conversion and punctuation strip.

Core ML Algorithms

TF-IDF Vectorization

Term Frequency-Inverse Document Frequency

Evaluates word importance mathematically, downweighting common words and prioritizing domain-specific skills (e.g., "Python", "GAAP").

Classification Model

Random Forest / SVM / KNN

Supervised learning model trained on the vectorized matrix to define decision boundaries separating distinct job categories.

MODEL PERFORMANCE

Testing Metrics

98%

Global Accuracy

0.97

Macro F1-Score

Challenges & Limitations

Parsing Challenges

Complex Layouts Multi-column resumes confuse standard top-down PDF parsers.

Implicit Skills TF-IDF misses contextual skills not explicitly named in the text.

Future Improvements

Word Embeddings Transitioning from TF-IDF to Word2Vec or BERT for semantic context.

Named Entity Recognition (NER) Extracting exact entities (Degree, Org, Years of Exp).

Technical Q&A

1. TF-IDF vs Word Embeddings (Word2Vec)?

TF-IDF represents term frequency but ignores semantic meaning and word relationships.

2. How to handle class imbalance in HR data?

Use SMOTE to oversample minority classes or adjust class weights in the classifier.

3. Why remove stopwords?

Reduces dimensionality and noise, allowing the model to focus on domain-specific keywords.

4. Precision vs. Recall in resume screening?

High recall is preferred so no good candidate is missed, leaving final filtering to humans.

5. Dealing with Out-Of-Vocabulary (OOV) words?

Subword tokenization (like BPE used in BERT) handles OOV by breaking them into known chunks.

Data-Driven Talent Discovery.

Scaling HR operations through intelligent natural language processing.



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Image Sources



https://repository-images.githubusercontent.com/396035212/8b111666-419f-431f-bef9-1dcfda6efcd0

Source: github.com



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